



# Weichi Zhao

## AI Infrastructure / LLM Serving Platform Engineer

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Target roles: LLM serving platform, OpenAI-compatible MaaS, AI gateway, agent platform, heterogeneous GPU/XPU scheduling.

### Profile

AI infrastructure engineer focused on production LLM serving platforms and model-service control planes. I build OpenAI-compatible MaaS capabilities, public model services, offline batch inference, model gateways, agent-client integration, routing/orchestration reliability, and Kubernetes-native serving on heterogeneous GPU/XPU clusters. In current MaaS and LLM serving work, I embed Claude Code, Codex, OpenCode, OpenClaw, and similar agent tools into code search, debugging, documentation, release checks, and verification loops to iterate faster on complex platform changes.

### Impact Highlights

- Scaled BatchFlow for production data distillation and batch generation workloads, later supporting around 10M requests per day with success rate above 99%.
- Validated BatchFlow -> SGLang Router -> multi-KLX backend for Qwen3: about 300K requests / 9.75B tokens, about 99.9% success rate, and below 5% BatchFlow/Router CPU and memory overhead.
- Built OpenAI-compatible MaaS public-service capabilities around API keys, model access, TPM/RPM limits, token budgets, model mapping, and reusable onboarding flows for agent clients.
- Applied agent-assisted workflows in current MaaS/LLM serving work for code search, incident writeups, design drafts, release checks, and knowledge-base updates, turning repeated manual steps into reusable iteration loops.
- Hardened serving reliability across gateway, Router, Worker, Service discovery, readiness, and long-connection failure modes instead of relying only on Pod-level health.

### Selected IP & Honors

- Co-inventor on granted invention patents for elastic model inference serving and cache-aware deep-learning training.
- Contributor to authorized software copyrights for cloud/HPC workflow platforms; team-related honors include China Youth May Fourth Medal collective honor, National Worker Pioneer recognition, and Gordon Bell Prize team context.

### Core Skills

|               |                                                                                                                                                  |                     |                                                                                                                                                                                     |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LLM Serving   | SGLang, vLLM, OpenAI-compatible APIs, streaming, embeddings, Responses API, PD disaggregation, KV/prefix cache, cache-aware and bucket routing.  | MaaS Public Service | Higress, LiteLLM, AI Gateway, API keys, team/model access, TPM/RPM limits, token budgets, model mapping, observability.                                                             |
| Control Plane | Kubernetes operators/controllers, CRDs, Helm, Kustomize, readiness, finalizers, service discovery, rollout safety, retry and rollback workflows. | Agent Platform      | Claude Code / Codex / OpenCode integration, tool-calling API compatibility, OpenAI/Anthropic-style adapters, long-context cache optimization, agent-assisted engineering iteration. |
|               |                                                                                                                                                  | Engineering         | Go, Python, Linux, Prometheus metrics, MySQL/SQLite/Redis, distributed systems debugging, performance and capacity testing.                                                         |

### Experience

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| <b>Zhejiang Lab, High-Performance Computing Facility Research Center</b><br>Senior Research Specialist, AI infrastructure and MaaS platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2024 - Present  |
| <ul style="list-style-type: none"><li>Built and operated LLM serving platform capabilities across CPU ACK and GPU ACK clusters, covering public model-service endpoints, OpenAI-compatible APIs, Higress/LiteLLM gateways, SGLang Router based model routing, IGD/IGDSA inference orchestration, BatchFlow offline inference, and Agent-facing API compatibility.</li><li>Owned production troubleshooting and hardening for LLM serving paths, including Router worker registration convergence, Service/Endpoint readiness, PD prefill/decode topology, CrashLoop and long-connection failures, gateway header/model mapping issues, and multi-cluster rollout validation.</li><li>Designed MaaS-Controller and optimizer-loop mechanisms around LiteLLM, Router, IGDSA, BatchFlow, and gateway metrics, with public-model limit adjustment, offline concurrency tuning, observation windows, rollback criteria, and online-SLA-first rules.</li></ul> |                 |
| <b>Zhejiang Lab, Intelligent Supercomputing Research Center</b><br>Engineering Specialist, cloud-native AI and HPC platforms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Aug 2021 - 2023 |
| <ul style="list-style-type: none"><li>Worked on cloud-native AI/HPC platform projects, including Kubernetes adaptation on domestic OS and accelerators, device plugin customization for accelerator availability, PyTorch/model adaptation, and distributed training framework exploration.</li><li>Supported model training and inference platform adaptation in domestic heterogeneous hardware/software environments, covering job submission, resource discovery, container runtime validation, and baseline performance verification.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                     |                 |

### Selected AI Infrastructure Projects

|                                                                                    |                     |
|------------------------------------------------------------------------------------|---------------------|
| <b>BatchFlow - OpenAI-compatible offline inference platform</b>                    | LLM batch inference |
| Asynchronous JSONL batch execution for large-scale, non-real-time model workloads. |                     |

- Designed Batch API compatible capabilities: file upload, batch lifecycle, cancellation, result/error files, streaming file processing, request-level error isolation, and backend concurrency control.
- Refactored BatchFlow toward multi-replica operation with Kubernetes leader election, sharded workers, header forwarding, upload limits, stale-file cleanup, Kustomize upgrades, monitoring, and failure-tolerant finalization.
- Supported production data-distillation and batch-generation workloads, later reaching daily processing around 10M requests with success rate above 99% under controlled concurrency.
- Validated a BatchFlow -> SGLang Router -> multi-KLX path for Qwen3-235B-A22B-Thinking-2507 at about 300K requests, about 9.75B tokens, about 99.9% success rate, and below 5% BatchFlow/Router CPU/memory overhead.

#### SGLang Router, IGD, and production serving reliability

#### Routing and orchestration

Reliable worker registration, routing, and recovery for multi-worker and PD-disaggregated inference.

- Improved Router worker synchronization by reconciling desired workers against live `/workers` state and requeueing on missing, extra, or failed registrations.
- Enhanced IGD/ISI startup and readiness with Pod-IP early registration, `publishNotReadyAddresses`, preserved Service finalizers, Router metrics Service generation, and ISI-level auto-restart with retry budgets.
- Led production RCA for ARG\_MAX overflow from service-link environment variables, Router not managed after Worker Ready, stale Service selectors, and long-connection timeout behavior.

#### MaaS public model service and agent integration optimization

#### OpenAI-compatible API / Agent runtime

Unified model access, authentication, rate limiting, protocol compatibility, cost governance, and troubleshooting for internal users, agent clients, and offline workloads.

- Built OpenAI-compatible public model-service endpoints covering Chat Completions, Responses-style APIs, embeddings, Batch API, and agent-client access while hiding differences across model backends, routers, and heterogeneous clusters.
- Contributed public-service governance around API keys, teams, model access, quotas, TPM/RPM limits, token budgets, model mapping, and cost-control policies for shared model access.
- Analyzed protocol compatibility boundaries for Claude Code / Codex-style agent clients, including system prompts, tool schemas, tool results, streaming events, Responses-style APIs, and OpenAI/Anthropic field differences.
- Diagnosed low SGLang prefix/KV cache hit rates for Claude Code long-context requests by comparing rendered prompts, cached-token logs, and token longest common prefix; identified dynamic attribution fields that polluted request prefixes and mitigated the issue.
- Embedded agent tools into current engineering workflows for cross-repository code navigation, log summarization, design drafts, PR/issue writing, release checks, and knowledge-base updates, shortening feedback loops for complex platform tasks.
- Established an agent-to-MaaS troubleshooting path across API keys, model permissions, quotas, max\_tokens, rate limits, connection timeouts, gateway forwarding, Router/Worker state, and model protocol behavior.

#### Kubernetes device plugin and heterogeneous accelerator platform

#### GPU/XPU scheduling

Cloud-native resource discovery, scheduling, and validation for domestic heterogeneous accelerator environments.

- Customized device-plugin and resource-management paths for accelerator availability, node resource reporting, container runtime validation, card-level scheduling, and failure isolation.
- Built deployment, upgrade, rollback, and E2E validation procedures for Kubernetes AI/HPC environments on domestic operating systems and accelerator stacks.

### Earlier Projects

#### Arizona State University, SNAC Lab

Jan 2020 - Aug 2020

Virtual platform developer and research assistant

- Developed modules for THOTH Lab, an open virtual lab platform for education and research, including SDN/OpenFlow lab modules and AWS instance control through Boto3-backed REST APIs.
- Implemented an Android prototype for an IR-CP-ABE based network security research system.

#### Arizona State University, CIDSE

Aug 2019 - Dec 2020

Graduate service assistant

- Assisted CSE205 Object-Oriented Programming, CSE412 Database Management, CSE434 Computer Networks, and CSE464 Software QA and Testing.

### Education

#### Arizona State University

May 2019 - Dec 2020

Master of Computer Science, GPA 3.97/4.00

#### Arizona State University

Aug 2015 - May 2019

Bachelor of Science in Computer Science, GPA 3.40/4.00